

DATABASE_SCHEMA.md

SynChain Database Schema

Database Choice

Current & Production

PostgreSQL

Reason:

- Open Source
 - Free
 - Production Ready
 - Industry Standard
 - Supports Analytics
-

users

Stores platform users.

Fields:

- id
 - name
 - email
 - password_hash
 - role
 - created_at
-

companies

Stores organizations.

Fields:

- id
- company_name
- industry

- country
 - created_at
-

products

Stores product information.

Fields:

- id
 - company_id
 - product_name
 - category
 - sku
 - created_at
-

warehouses

Stores warehouse information.

Fields:

- id
 - company_id
 - warehouse_name
 - location
 - capacity
 - created_at
-

suppliers

Stores supplier information.

Fields:

- id
- supplier_name
- location
- reliability_score
- created_at

simulations

Stores every simulation run.

Fields:

- id
- user_id
- product_id
- current_stock
- demand
- supplier_delay
- created_at

digital_twin_states

Stores virtual supply chain state.

Fields:

- id
- simulation_id
- inventory_state
- demand_state
- warehouse_state
- supplier_state
- risk_state
- created_at

agent_outputs

Stores outputs from every agent.

Fields:

- id
- simulation_id
- agent_name
- output_json
- confidence_score

- created_at
-

recommendations

Stores final recommendations.

Fields:

- id
 - simulation_id
 - recommendation
 - explanation
 - priority
 - created_at
-

news_signals

Stores external intelligence.

Fields:

- id
 - title
 - source
 - category
 - impact_score
 - published_at
-

risk_events

Stores identified risks.

Fields:

- id
- simulation_id
- risk_type
- severity
- description
- created_at